

Vijayan V

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PROFESSIONAL SUMMARY

B.Tech Electronics and Communication Engineering (ECE) graduate with hands-on experience in semiconductor device simulation using Silvaco ATLAS and a strong foundation in semiconductor devices, fabrication, and process integration. First-author IEEE-published researcher with experience in semiconductor device modeling, simulation, and performance analysis. Familiar with thin-film deposition, etching, cleanroom practices, and semiconductor manufacturing concepts through academic coursework and research. Passionate about semiconductor technology and eager to contribute to device simulation, process integration, fabrication, and advanced semiconductor R&D.;

EDUCATION

B.Tech — Electronics and Communication Engineering

May 2026

SRM Institute of Science and Technology, Kattankulathur, Chennai | CGPA: 8.17 / 10

RELEVANT EXPERIENCE

Defect-Tolerance Analysis of ZnO Nanorod Electron Transport Layers in Perovskite Solar Cells — First Author

Jan 2026 – May 2026

SRM Institute of Science and Technology | Under supervision of Dr. Md Jawaid Alam

- Research paper accepted at 2026 IEEE IPRECON as first author, on a comparative TCAD study of planar and nano-structured ZnO ETLs for perovskite solar cells
- Designed and simulated ZnO-based perovskite solar cell device (ITO/ZnO/MAPbI₃/CuI/Au stack) in Silvaco ATLAS — analyzed band alignment, carrier transport, interface defect effects, and J-V characteristics
- Compared planar vs. nanorod ZnO ETL architectures; showed nano-structured ETL achieves 26% power conversion efficiency vs. 22% for planar ETL in a defect-free environment, driven by higher interfacial area, vertical charge transport channels, increased short-circuit current density, and reduced recombination loss

TFET Device Simulation Research

June 2026 – Ongoing

SRM Institute of Science and Technology | Under supervision of Dr. Md Jawaid Alam

- Simulating AlN/GaN-based Tunnel FET structures in Silvaco ATLAS — investigating device electrostatics, carrier transport, and switching characteristics
 - Achieved electric field match of ~11 MV/cm against published paper's 12 MV/cm; extracted Ion/Ioff ratio of 1.57×10^{11} for double gate configuration
 - Built automated iteration loops in Silvaco ATLAS to sweep device parameters across multiple design points, followed by data analysis to identify the optimized device configuration
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TRAINING & CERTIFICATIONS

Semiconductor Technology — Summer Training & Internship Program

June - July 2026

E&ICT; Academy, IIT Roorkee | MeitY & IIT Roorkee Joint Initiative

- Hands-on TCAD-based training in Microelectronics, CMOS Technology, Wafer Fabrication, Oxidation, Diffusion, Implantation, Lithography, and Etching using Synopsys Sentaurus and Siemens Mentor
- Completed three hands-on TCAD projects: Si and Ge PN junction diode simulation with STI, 180nm 2D/3D NMOS design, and NMOS characterization with CMOS inverter performance analysis

India Semiconductor Workforce Development Program (ISWDP)

June 2026

IISc Bangalore | In collaboration with Synopsys and Samsung Semiconductor

- Completed Level 1 — TCAD device simulation workflow using Synopsys Sentaurus Device Editor, Sentaurus Device, and Sentaurus Workbench; diode and MOSFET simulation and characterization

IISc Bangalore — CeNSE Winter School

December 2025

Indian Institute of Science, Bangalore | Certificate of Distinction

- Completed intensive 2-week program covering full semiconductor fabrication flow: wafer cleaning, doping, thin film deposition (CVD/PVD/ALD), photolithography, plasma etching, electrical characterization, SEM/TEM microscopy, MEMS, advanced packaging, and statistical process control
- Awarded Certificate of Distinction based on technical evaluation — recognized among top performers

Semiconductor Process Integration — Coursework and Lab

2024–2026

SRM IST | In collaboration with IISc CeNSE, Lam Research, India Semiconductor Mission

- Completed 12-week intensive course covering complete CMOS process flow: lithography (UV/DUV/EUV/SADP/SAQP), CVD/PVD/ALD, dry and wet etching, CMP, ion implantation, gate stack engineering (RMG/MIPS/HfO2 ALD), BEOL metallization, Damascene/dual-Damascene, and TSV
- Executed 14-nm FinFET process emulation in SEMulator3D; independently designed 3D NAND (Inspired by Micron Technology) and monolithic IC process flows — each involving 100+ fabrication steps
- Achieved Grade O (Outstanding) — the highest grade available — and was recognized by faculty for outstanding mini project; selected as Student Mentor — Semiconductor Design Club

Fundamentals of Micro and Nanofabrication — NPTEL

IISc Bangalore | Prof. Sushobhan Avasthi and Prof. Shankar Kumar Selvaraja | Certificate of Completion

- Cleanroom equipment protocols: mask aligner, four-point probe, spin coater, thermal evaporator, sputtering system, wet bench operations; photolithography, thin film deposition, and device characterization methods

TECHNICAL SKILLS AND CONCEPTUAL KNOWLEDGE

TCAD & Simulation	Silvaco ATLAS (device simulation) · SEMulator3D (process modeling, 3D visualization)
Semiconductor Process	FEOL/BEOL integration · Lithography (UV, DUV, EUV, SADP/SAQP, multiple patterning) · CVD/PVD/ALD · Dry/wet etching · Ion implantation · CMP · Gate stack engineering (RMG, MIPS, HfO2) · Damascene/dual-Damascene · TSV · FinFET process emulation
Device Physics	PN junctions · MOSFETs (short channel effects, threshold voltage) · MOS capacitor · BJT · Carrier transport (drift, diffusion, recombination) · High-k dielectrics · TFET fundamentals · Optoelectronic devices
Data Analysis & Programming	Python · NumPy · Pandas · Matplotlib · MATLAB · Google Colab · Jupyter Notebook · SQL · Power BI
Documentation	LaTeX · Overleaf · Microsoft Office Word · Excel · Powerpoint

KEY ACHIEVEMENTS

- First author — "Defect-Tolerance Analysis of ZnO Nanorod Electron Transport Layers in Perovskite Solar Cells," accepted at 2026 IEEE IPRECON
- Certificate of Distinction — IISc CeNSE Winter School 2025 (top performer recognition)
- Grade O (Outstanding) in Semiconductor Process Integration course — highest grade available
- Designed 3D NAND and monolithic IC process flows in SEMulator3D — 100+ steps each — at undergraduate level
- Student Mentor — Semiconductor Design Club, SRM IST — mentored juniors across three consecutive semesters

COMMUNICATION

- English - Professional Level
- Tamil - Native
- Mandarin Chinese - Basic